

Overall Performance Comparison (Mean $\pm$ 95% CI)

	Total Energy (kJ) ↓	High-Priority Satisfaction (%) ↑	Medium-Priority Satisfaction (%) ↑	Low-Priority Satisfaction (%) ↑
2D-AUTO (47 m)	81.5 $\pm$ 0.8	0.8 $\pm$ 1.8	61.6 $\pm$ 8.3	68.0 $\pm$ 6.5
Two-Stage (decoupled)	79.4 $\pm$ 1.5 *	55.0 $\pm$ 6.9 ***	73.2 $\pm$ 7.2 *	53.5 $\pm$ 6.8 ***
Coupled-Greedy (ablation)	79.4 $\pm$ 2.1 n.s.	51.6 $\pm$ 8.5 ***	69.4 $\pm$ 8.0 n.s.	71.1 $\pm$ 9.3 n.s.
Strong-Coupled (deterministic)	78.1 $\pm$ 2.0 **	67.4 $\pm$ 8.6 ***	78.2 $\pm$ 4.9 **	75.7 $\pm$ 11.2 n.s.

Bold marks the best value per column.

*Markers: paired t-test vs 2D-AUTO — *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant.*

$n = 10$ city seeds; $\pm$ is the Student-t 95% CI.

Ours vs Two-Stage (decoupled), paired: energy $p = 0.245$, high $p = 0.033$, medium $p = 0.322$, low $p = 0.001$.

Means match Figs. 5-6; ↓ lower is better, ↑ higher is better.